
Collective Semantic Memory: Merge, Trust, and Staleness for Peer Memory in Multi-Agent Navigation

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Abstract

1 **Problem.** Collective memory is often treated as monotone: more shared information should help. In cooperative navigation with scarce semantic targets, this
2 intuition fails. Fast global sharing improves target materialization but causes agents
3 to race for the same targets; slow sharing avoids contention but leaves agents with
4 stale or incomplete memories. **Approach.** We instantiate four memory topologies
5 over the same symbolic-semantic substrate: independent private memories, a
6 shared event bus, a central consensus aggregator, and peer-to-peer broadcast with
7 cadence K . We then introduce a minimal *collective semantic memory* (CSM): peer
8 broadcast plus explicit *merge*, *trust*, and *staleness* rules. Q/R/M/C stage rates are
9 used only as measurement instrumentation, not as the contribution of this paper.
10 **Findings.** On a 35,640-run multi-agent sweep, the fixed-cadence peer family
11 exhibits a bottleneck shift: fast broadcast saturates materialization but collapses
12 completion, while slow broadcast has the opposite profile. Mean time-to-success
13 has an interior optimum at $K=8$. On a 3,240-run scarcity benchmark, the minimal
14 CSM strictly dominates five fixed-cadence peer baselines on success rate: CSM
15 reaches 0.617 with 95% CI [0.610, 0.624], above every peer-K upper CI. **Scope.**
16 Experiments are grid-world substrates. The claim is not that this is a complete
17 theory of collective memory; it is a reproducible architecture split showing that
18 merge/trust/staleness are the right explicit control surfaces after cadence alone is
19 exhausted.
20

21 1 Introduction

22 Many multi-agent systems share information either by writing to a single global memory, by sending
23 learned messages, or by periodically exchanging local state. These mechanisms are usually evaluated
24 end-to-end. If success improves, communication is judged useful; if it does not, the reason is opaque.
25 This paper studies a narrower architectural question: when multiple agents maintain semantic memory
26 about places, what should be shared, at what rate, and with what weight?

27 The answer is not “share everything as soon as possible.” In a scarcity setting, where N agents
28 compete for $T < N$ semantic targets, fresh global memory can make all agents select the same
29 target. The memory is then accurate but unhelpful. Conversely, no communication prevents herding
30 but starves agents of useful discoveries made by peers. The design problem is therefore a trade-off
31 between semantic freshness and occupancy contention.

32 We make three contributions.

- 33 1. **Architecture split.** We implement and compare four memory topologies over the same symbolic
34 substrate: independent, shared-bus, central-aggregator, and peer-to-peer broadcast.

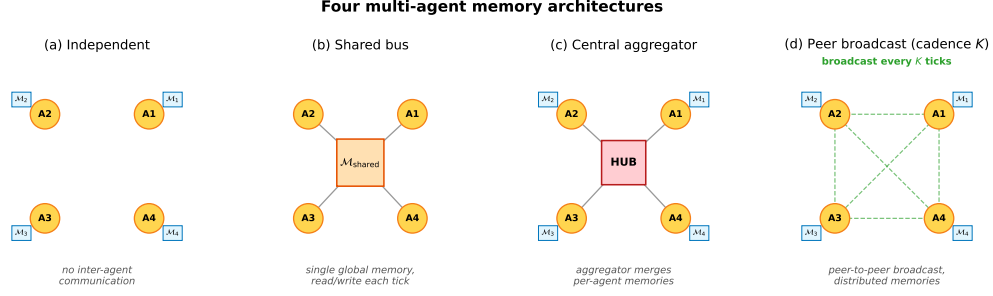


Figure 1: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

- 35 **2. Cadence result.** We show that peer broadcast exposes a real coordination parameter K : fast
 36 broadcast improves $P(M^*|R^*)$ but hurts $P(C^*|M^*)$; the efficiency metric has an interior
 37 optimum.
- 38 **3. Minimal CSM.** We instantiate collective semantic memory with three explicit rules: merge,
 39 trust, and staleness. The minimal CSM reaches a new point on the $M \times C$ plane and strictly
 40 dominates fixed-cadence peer baselines on success rate in the scarcity benchmark.
- 41 **Relation to the companion Q/R/M/C paper.** The companion paper treats Q/R/M/C as a diagnostic
 42 framework. Here those stage rates are used as measuring instruments. The main object is different:
 43 the design of distributed symbolic memory itself.

44 2 Memory substrate and task

45 Each agent observes a grid environment and emits semantic place evidence such as `water_source`,
 46 `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts,
 47 freshness, and transition statistics. Planner queries retrieve candidate places by matching required,
 48 preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by
 49 a local controller.

50 The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets
 51 are scarce: $N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts, three hazard densities, and 40 seeds for
 52 the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*,$$

53 where M^* means a semantic candidate was locked as an actionable target and C^* means the target
 54 was reached. These logs are not the architectural contribution; they let us see where each topology
 55 fails.

56 3 Four collective-memory topologies

57 The four topologies share the same local memory and planner. They differ only in where merge
 58 happens.

59 **Independent.** Each agent owns a private event store and concept graph. There is no communication
 60 path.

61 **Shared bus.** All observations are appended to a single global event bus. One shared concept graph
 62 is rebuilt from the union of all events.

63 **Central aggregator.** Each agent owns a private concept graph; a central consensus layer pulls
 64 snapshots and computes one trust-weighted report for everyone.

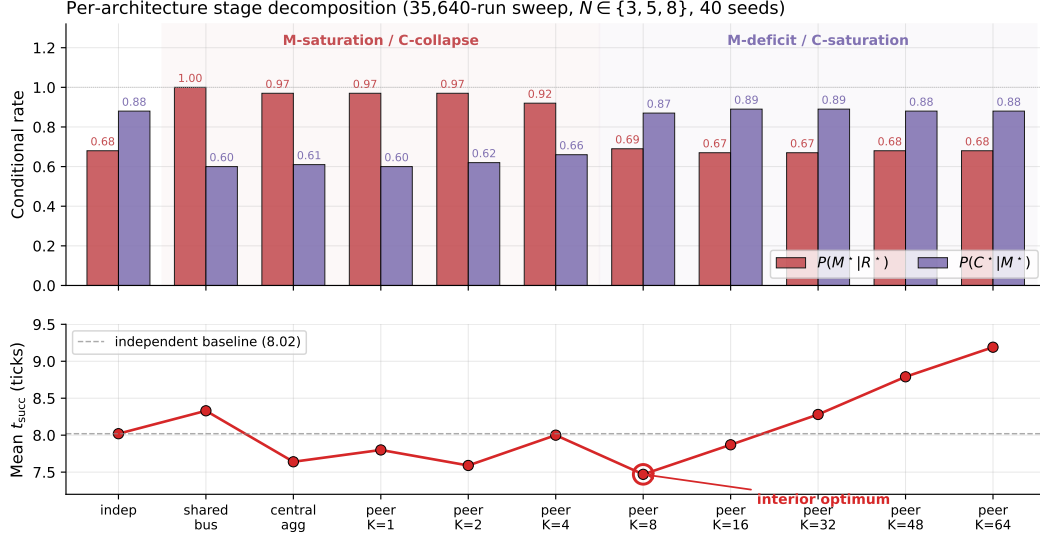


Figure 2: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Mean time-to-success has an interior optimum at $K=8$.

65 **Peer broadcast.** Each agent owns a private concept graph and a private merged view. Every K ticks,
 66 agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each
 67 peer.

68 The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

independent : N graphs, 0 mergers
 shared bus : 1 graph, 1 implicit global merger
 central : N graphs + 1 central merger
 peer : N graphs + N private mergers.

69 4 Cadence exposes the failure mode

70 The peer topology has one explicit parameter: broadcast cadence K . The main sweep crosses
 71 four architectures and eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 2
 72 summarizes the result.

73 Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can
 74 lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow
 75 peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but
 76 those that do are less likely to collide with peer occupancy.

77 Across all peer runs, Spearman of $P(M^* | R^*)$ vs K is -0.608 and of $P(C^* | M^*)$ vs K is $+0.420$
 78 (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson
 79 correlation between the M and C links peaks at -0.61 for $K=4$ and decays toward noise as K
 80 grows. Cadence therefore does not simply tune “amount of communication.” It moves the bottleneck
 81 between semantic commitment and shared-resource completion.

82 5 Minimal collective semantic memory

83 The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A
 84 collective semantic memory should expose at least three:

85 **Merge.** How peer evidence about the same semantic place is combined.

86 **Trust.** How evidence from different peers is weighted when it conflicts or becomes unreliable.

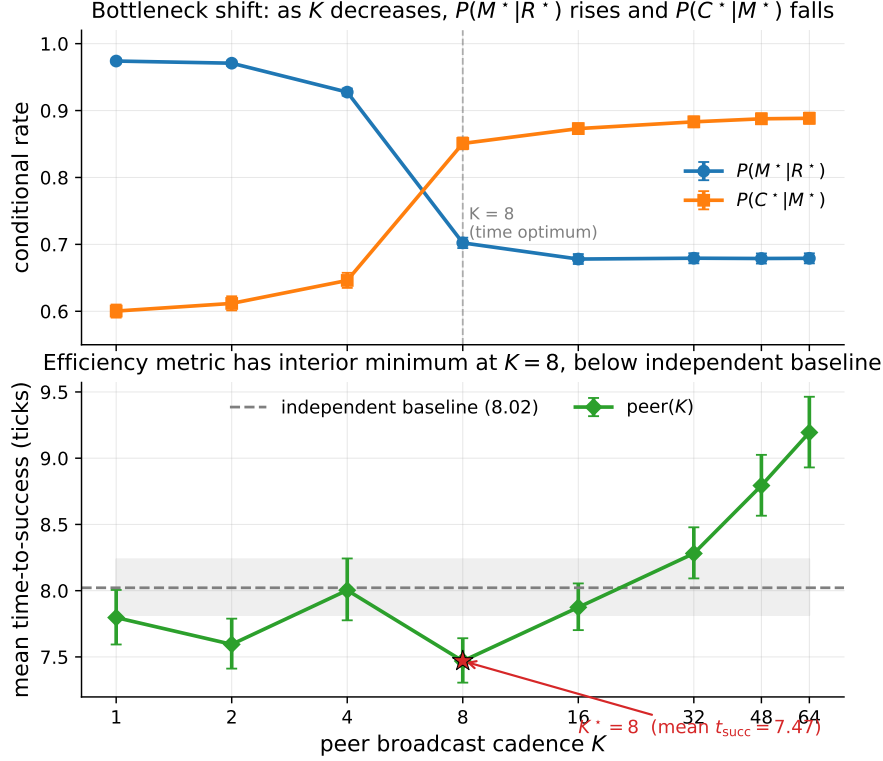


Figure 3: Bottleneck shift across peer cadences. $P(M^*|R^*)$ falls and $P(C^*|M^*)$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at $K=8$.

87 **Staleness.** How old evidence decays before it influences retrieval.

88 The minimal CSM uses peer broadcast at $K=8$ and overlays three rules:

89 **Merge.** Per-place score equals own evidence at weight 1 plus peer evidence at weight $\text{trust}_i \cdot e^{-\alpha \cdot \text{age}}$,
 90 with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

91 **Trust.** Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the
 92 locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

93 **Staleness.** Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha \cdot \text{age}}$. Snapshots
 94 older than $10K$ ticks are pruned.

95 The implementation is a drop-in replacement for fixed-cadence peer memory:
 96 `experiments/collective_semantic_memory/csm_memory.py`.

97 5.1 Results

98 We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells
 99 $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

100 CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion;
 101 slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute
 102 success gain is modest, +1.7 to +4.0 percentage points, but the stage decomposition shows why it
 103 is meaningful: the architecture changes the joint $M \times C$ operating point rather than merely moving
 104 along the fixed-cadence curve.

Table 1: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM’s lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^* R^*)$	$P(C^* M^*)$	Success rate (95% CI)
peer ($K=1$)	0.991	0.586	0.577 [0.572, 0.586]
peer ($K=4$)	0.955	0.623	0.581 [0.572, 0.590]
peer ($K=8$)	0.730	0.871	0.599 [0.591, 0.607]
peer ($K=16$)	0.697	0.892	0.597 [0.589, 0.606]
peer ($K=64$)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	0.782	0.825	0.617 [0.610, 0.624]

6 Categorical BDI reading of collective semantic memory

This section gives a compact theoretical reading of the water-search example in terms of belief, desire, and intention. The goal is not to introduce a new planner, but to make explicit the mathematical structure already used by the collective semantic memory architecture.

We assume a finite set of agents $A = \{1, \dots, N\}$, a shared spatial basis X , and a shared semantic alphabet Σ . In the present grid-world implementation, X is the common coordinate frame and Σ contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an explicit modelling assumption. The agents may have different observations, different memories, and different confidence values, but they interpret places in a common coordinate framework and express semantic evidence in a common tag alphabet. Without these two assumptions, peer merge would require a separate place-alignment and symbol-alignment problem, which is outside the scope of the current experiment.

At time t , agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place records stored in the agent’s private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation I_t^i is the thresholded skeleton needed for the theoretical description.

This formal context induces a Galois connection between sets of tags and sets of places:

$$\text{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B, (g, \sigma) \in I_t^i\}, \quad \text{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S, (g, \sigma) \in I_t^i\}.$$

For a water-search task, the desired semantic predicate is $B_{\text{water}} = \{\text{water_source}\}$, or, in the implemented weighted form, a predicate combining positive evidence for `water_source`, `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval succeeds for agent i at time t exactly when

$$\text{ext}_t^i(B_{\text{water}}) \neq \emptyset.$$

Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning already has a non-empty extent in the agent’s realized memory.

We can now define the BDI components.

Belief. The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it is the concept lattice $L_t^i = \text{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags. A water belief is not the proposition “there is water at coordinate x .” It is the region of L_t^i whose intension contains the water predicate. If this region is empty, the agent does not merely choose badly; the relevant semantic object is absent from its current memory.

Desire. Desire is a state-conditioned semantic predicate over the belief lattice. In the water example, the agent’s internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\text{water}}, & \text{if } \text{thirst}_t^i \geq \theta_{\text{on}}, \\ B_{\text{goal}}, & \text{otherwise.} \end{cases}$$

Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate should be queried from the current belief structure.

139 **Intention.** Intention is the commitment to one materialized candidate. Once desire has selected
 140 a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice
 141 function

$$\chi_t^i : \mathcal{P}(G_t^i) \rightarrow G_t^i \cup \{\perp\}, \quad g_t^{i,*} = \chi_t^i(\text{ext}_t^i(D_t^i)),$$

142 where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query Q^* ,
 143 belief supports retrieval R^* , and intention corresponds to materialization M^* . Completion C^* is
 144 downstream of this semantic structure: it depends on whether the physical controller reaches the
 145 locked target and whether the target remains available under multi-agent occupancy.

146 This gives a compact categorical reading. Let $\mathbf{Ctx}_\Sigma$ be the category whose objects are semantic
 147 contexts (G, Σ, I) over the fixed alphabet Σ , and whose morphisms preserve semantic incidence. A
 148 local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \rightarrow \mathcal{K}_1^i \rightarrow \dots \rightarrow \mathcal{K}_t^i \rightarrow \mathcal{K}_{t+1}^i \rightarrow \dots$$

149 in $\mathbf{Ctx}_\Sigma$. Each arrow is induced by observation and memory update. When new evidence is
 150 consolidated, the formal context changes; therefore its concept lattice changes. This is the order-
 151 theoretic content of memory growth.

152 The H-MDP view is obtained by treating the high-level state of agent i as

$$h_t^i = (L_t^i, D_t^i, g_t^{i,*}, s_t^i),$$

153 where L_t^i is the belief lattice, D_t^i is the active desire predicate, $g_t^{i,*}$ is the current intention, and s_t^i
 154 is the low-level physical state, including position, risk budget, thirst, and energy. The transition
 155 $h_t^i \rightarrow h_{t+1}^i$ combines two coupled processes: a semantic-memory update $\mathcal{K}_t^i \rightarrow \mathcal{K}_{t+1}^i$ and a physical
 156 control transition $s_t^i \rightarrow s_{t+1}^i$. The useful point is that the semantic part and the physical part fail in
 157 different ways. An empty extent $\text{ext}_t^i(D_t^i) = \emptyset$ is a belief/retrieval failure. A non-empty extent with
 158 no stable lock is a materialization failure. A successful lock that cannot be reached is a completion
 159 failure.

160 We now lift the construction to the collective case. At broadcast times, agent i receives snapshots
 161 from peers. For the receiver i , these snapshots form a diagram of local memory objects

$$\mathcal{D}_t^i = (\mathcal{K}_t^i, \mathcal{K}_t^{j_1}, \dots, \mathcal{K}_t^{j_m}),$$

162 together with alignment maps induced by the shared coordinate basis and semantic tag compatibility.
 163 The merged view used by agent i is a colimit-like consensus object

$$\tilde{\mathcal{K}}_t^i = \text{Merge}_{\text{trust, stale}}(\mathcal{D}_t^i).$$

164 This notation is intentionally modest. The implementation does not claim to solve general semantic
 165 alignment. It computes a practical merge over aligned place records, with peer evidence weighted by
 166 trust and staleness. A typical weight has the form

$$w_{j \rightarrow i}(t) = \text{trust}_{j \rightarrow i}(t) \exp(-\alpha \text{age}_j(t)),$$

167 so the merged incidence strength for a tag σ at a place g can be written schematically as

$$\tilde{I}_t^i(g, \sigma) = I_t^i(g, \sigma) + \sum_{j \neq i} w_{j \rightarrow i}(t) I_t^j(g, \sigma),$$

168 followed by thresholding or weighted scoring during retrieval. This separates three control surfaces:
 169 *merge* (what evidence refers to the same place), *trust* (whose evidence should count more), and
 170 *staleness* (how quickly old evidence loses force).

171 The same BDI definitions then apply to the merged context $\tilde{\mathcal{K}}_t^i$. The agent's collective belief is
 172 the lattice $\tilde{L}_t^i = \text{Concepts}(\tilde{\mathcal{K}}_t^i)$, its desire is still a semantic predicate D_t^i , and its intention is now
 173 selected from $\text{ext}_{\tilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from its private extent alone.

174 This explains why communication cadence produces a bottleneck shift. Faster broadcast increases
 175 the freshness of peer evidence. This tends to enlarge or improve the water extent $\text{ext}_{\tilde{\mathcal{K}}_t^i}(B_{\text{water}})$, so
 176 agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^* \mid R^*)$. But the
 177 same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce

178 water source. The memory is semantically accurate, but the physical target is now contested. This
 179 lowers $P(C^* \mid M^*)$. The bottleneck is therefore physically grounded. It is not a formal artifact of
 180 the decomposition: it arises because semantic belief is shared through memory, while completion is
 181 constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate
 182 not only how much information is shared, but also how it is merged, whose evidence is trusted, and
 183 when stale evidence should stop influencing intention formation.

184 In this sense the “collective” is not a central mind and not a global map but a stabilised family of
 185 peer-indexed processes; the next subsection makes that statement precise.

186 6.1 The coalgebraic layer: individuals as codata, the collective as a final coalgebra

187 The reading of §6 fixes belief *content*: at a state, the concept lattice $L_t^i = \text{Concepts}(\mathcal{K}_t^i)$ says what
 188 is retrievable, and the merge of §6.2 says how peer contents combine into $\tilde{\mathcal{K}}_t^i$. Both are *algebraic*:
 189 they build a belief object. Neither speaks to the agent as a non-terminating *process* that must emit an
 190 action and a signal at every tick under a moving environment, nor to the sense in which a *population*
 191 comes to share a convention. That process side is *coalgebraic* (Rutten, 2000; Jacobs, 2016). We
 192 make it precise on the water-search example and show the two sides are dual, meeting inside a single
 193 state transition. (The belief/desire/intention vocabulary follows the BDI agent tradition (Rao and
 194 Georgeff, 1995), here given coalgebraic content.)

195 **Signature.** Fix an action set Act (move, dig, noop), a per-agent observation set O_i (agent i ’s local
 196 percept — its *unique view*), and an *ostension* set

$$M = 1 + (X \times \Sigma),$$

197 whose values are either *silence* (the summand 1) or a pointer “place x carries tag σ ” (whistle-and-point
 198 at a water-marked cell). That silence is a first-class value of M matters: the ostension channel *can be*
 199 *informatively empty*, and it is separate from the state update below. This is exactly the observability
 200 that a learned real-valued message vector destroys (its channel is never empty and never separates
 201 signalling from stepping), and it is why the symbolic stages remain readable.

202 **State.** We take the coalgebra carrier to be the H-MDP high-level state of §6,

$$\mathcal{S}_i = \{ h = (\mathcal{K}^i, D^i, g^{i,*}, s^i) \},$$

203 carrying the same information as $(L^i, D^i, g^{i,*}, s^i)$ since $L^i = \text{Concepts}(\mathcal{K}^i)$. Thus *belief lives in*
 204 *the state*; the lattice is a projection of it.

205 **Definition (codata: the agent’s cotype).** The observable behaviour of agent i is a coalgebra for the
 206 endofunctor

$$B_i(S) = \underbrace{\text{Act}}_{\text{act now}} \times \underbrace{M}_{\text{signal now}} \times \underbrace{S^{O_i \times M^{A \setminus \{i\}}}}_{\text{continue given (my next percept, peers’ signals)}},$$

207 i.e. a map $c_i = \langle \text{observe_act}_i, \text{observe_ostension}_i, \text{step_intension}_i \rangle : \mathcal{S}_i \rightarrow B_i(\mathcal{S}_i)$
 208 with three destructors

$$\text{observe_act}_i : \mathcal{S}_i \rightarrow \text{Act}, \quad \text{observe_ostension}_i : \mathcal{S}_i \rightarrow M, \quad \text{step_intension}_i : \mathcal{S}_i \rightarrow \mathcal{S}_i^{O_i \times M^{A \setminus \{i\}}}.$$

209 Here $M^{A \setminus \{i\}} = \prod_{j \neq i} M$ is the tuple of peers’ ostensions delivered this tick. Operationally,
 210 $\text{step_intension}_i(h)(o, (m_j)_{j \neq i})$ (i) folds the percept o and the peer pointers (m_j) into the context
 211 by the trust/staleness merge, $\tilde{\mathcal{K}}_t^i = \text{Merge}_{\text{trust, stale}}(\dots)$ of §6.2; (ii) recomputes the lattice; (iii) re-
 212 evaluates the desire predicate D^i ; and (iv) re-locks the intention $g^{i,*} = \chi_t^i(\text{ext}(D^i))$ of §6. So Merge
 213 and χ are *components of one destructor*. The agent *is* the codata generated by c_i : it has no finite
 214 terminal plan, only the standing ability to emit an action and an ostension and to continue. This is the
 215 “infinite survival generator” of the motivating example.

216 **Definition (indexed family).** The system is the A -indexed family $\{(\mathcal{S}_i, c_i)\}_{i \in A}$, a coalgebra in
 217 Set^A . The functor B_i depends on the index i through O_i (and through the peer set $A \setminus \{i\}$): the
 218 same survival cotype is instantiated at each agent through that agent’s local viewpoint. Agent 1 on
 219 the dune generates from “I see the horizon”; agent 2 in the hollow generates from “I feel the soil.”
 220 The index is the viewpoint.

221 **Definition (codependence: the collective coalgebra).** Let the environment deliver a joint percept
 222 $o = (o_i)_{i \in A} \in \prod_i O_i$. The collective one-tick map on $\prod_i \mathcal{S}_i$ wires each agent’s ostension output
 223 into the others’ step inputs,

$$\Phi((h_i)_i)(o) = \left(\text{step_intension}_i(h_i)(o_i, (\text{observe_ostension}_j(h_j))_{j \neq i}) \right)_{i \in A}.$$

224 Because i ’s next state is a function of the *destructor outputs* of the others, the family is not N
 225 independent processes but one coalgebra for the product functor $\mathbf{B}(S) = (\prod_i (\text{Act} \times M)) \times S \prod_i O_i$
 226 on $\prod_i \mathcal{S}_i$. This wiring is the *codependence*: agent 1 cannot step without passing agent 2’s ostension
 227 through its own `step_intension`. Broadcast cadence enters here as a gate: at non-broadcast ticks
 228 the delivered tuple (m_j) is the last received (stale) tuple, or silence; K is the delivery period.

229 **Definition (behaviour and bisimulation).** B_i is a container (polynomial) functor, so it has a final
 230 coalgebra $(\nu B_i, \text{beh}_i)$ (Aczel, 1988; Rutten, 2000) whose elements are the $(O_i \times M^{A \setminus \{i\}})$ -branching,
 231 $(\text{Act} \times M)$ -labelled behaviour trees, and there is a unique coalgebra morphism $!_i : \mathcal{S}_i \rightarrow \nu B_i$ sending
 232 a state to its whole future behaviour. A B_i -bisimulation is a relation $R \subseteq \mathcal{S}_i \times \mathcal{S}_i$ under which
 233 related states emit equal action and equal ostension and, on every input, step to R -related states.
 234 Container functors preserve weak pullbacks, so the *greatest bisimulation* $\sim_i$ exists, is an equivalence,
 235 and equals $\ker(!_i)$: two states are bisimilar iff they have the same behaviour (Sangiorgi, 2011). The
 236 same holds for the collective coalgebra $(\prod_i \mathcal{S}_i, \langle (\text{observe_act}_i)_i, (\text{observe_ostension}_i)_i, \Phi \rangle)$,
 237 with greatest bisimulation $\approx$.

238 **Definition (society / culture).** The *collective* is the image of the states in the final coalgebra $\nu \mathbf{B}$,
 239 equivalently the quotient $(\prod_i \mathcal{S}_i) / \approx$. It is neither a central server nor a shared map: it is a bisimilarity
 240 class of joint behaviour. A *water-search convention* is this invariant read on the water sub-signature.
 241 Let $M_{\text{water}} \subseteq M$ be the pointers whose tag lies in the water family, and let $\nu B_i \rightarrow \nu B^w$ be the
 242 forgetful map to the behaviour trees restricted to receiving M_{water} and emitting the corresponding
 243 intention lock. The population *holds the convention* when all agents’ reachable behaviours agree
 244 under this map: hearing “point-and-dig at a water-marked place” drives every agent’s intention
 245 destructor to lock that place (up to occupancy). Governance is then automatic in both directions. A
 246 new agent (the example’s fourth agent) is a member of the society exactly when its behaviour lands
 247 in the class $[\cdot]_{\approx}$ on M_{water} — otherwise its ostensions are not understood and its own step never
 248 converges (top-down); and the class itself was reached by the local, coupled steps of the first agents
 249 at one green shrub (bottom-up).

250 **Remark (communication = lazy, depth-one corecursion).** The behaviour $!_i(h)$ is defined by
 251 unfolding c_i indefinitely. On receiving $m_j = \text{observe_ostension}_j(h_j)$, agent i uses only this
 252 *depth-one* destructor output inside `step_intensioni`; it does *not* compute $!_j(h_j) \in \nu B_j$, the peer’s
 253 full behaviour, which is the “I think that you think...” regress. Communication is therefore corecur-
 254 sion truncated to the least depth that resolves the receiver’s own next step — a finite approximation
 255 of the final-coalgebra map — with a relevance-theoretic stopping rule (Sperber and Wilson, 1995)
 256 deciding that depth. The regress collapses into one effective action.

257 **Remark (algebra-coalgebra bridge to §6–6.2).** Belief is algebraic: the FCA lattice L_t^i built by
 258 the colimit merge is the *content at a state*. The agent is coalgebraic: the codata observed by the three
 259 destructors is the *process over states*. They are dual and meet in one place — the colimit merge
 260 and the satisficing choice χ are the internals of `step_intension`. The memory-growth diagram
 261 $\mathcal{K}_0^i \rightarrow \mathcal{K}_1^i \rightarrow \dots$ of §6 is precisely the belief-projection of the coalgebra trajectory $h_0^i \mapsto h_1^i \mapsto \dots$.

262 **Coalgebraic reading of the bottleneck shift.** The result of §4 and Table 1 has a one-line reading
 263 here. Two kinds of bisimilarity compete. Bisimilarity on the *ostension / belief* sub-signature is the
 264 wanted invariant: it is the shared water convention. Bisimilarity on the raw *intention* output over a
 265 *scarce* target is the unwanted one: several states mapping to the same g^* is a collision. Occupancy
 266 $\mathcal{O}_{-i}$ is the extra-coalgebraic constraint that penalises the second. Cadence K tunes how hard the
 267 coupling Φ drives the collective toward the *full* greatest bisimulation $\approx$: small K delivers fresh
 268 ostensions often, enlarging the merged water extent so intention locks reliably ($P(M^* \mid R^*) \uparrow$), but
 269 over-synchronises the intention destructor so agents collide ($P(C^* \mid M^*) \downarrow$). The interior optimum
 270 $K=8$ is where belief converges without intention collapsing onto one cell; the minimal CSM adds

Table 2: The motivating example’s terms as formal objects of the coalgebraic reading. Each row is the water-search meaning of a construction already used in §6.1.

Informal term	Formal object	Water-search meaning
Cotype (codata)	B_i -coalgebra (S_i, c_i)	the agent’s non-terminating action/signal generator
Indexed type	family over A ; dependence of B_i on O_i	the agent’s unique local viewpoint (dune vs. hollow)
Codependent type	wiring $\text{observe_ostension}_j \rightarrow \text{step_intension}_i$ in Φ	“I step only through your ostension”
Individual (instance)	a state $h \in S_i$ and its behaviour $!_i(h)$	the running agent digging and whistling now
Destructors (cogenerators)	observe_act , observe_ostension , step_intension	step-and-dig; whistle-and-point; fold neighbour’s signal into next state
Lazy evaluation	depth-one corecursion + relevance stop	unfold the peer’s coalgebra one step, not its whole behaviour
Final coinductive core / society	νB ; greatest bisimulation $\approx$ (on M_{water})	the stable, distributed water-search culture

trust and staleness precisely so that step_intension reaches belief-bisimilarity (fresh, trusted, aligned evidence) while damping intention-bisimilarity (stale evidence decays before it can herd every agent onto the same lock). We state this as a reading consistent with the data, not as a theorem: the container functors, their final coalgebras, and bisimulation are standard constructions used here as a lens, and the relevance-theoretic stopping rule is an interpretive gloss rather than a fitted mechanism.

6.2 Colimit merge and its invariants

The merge object above can be described more precisely. Working in a timestamp- and provenance-enriched version of $\mathbf{Ctx}_\Sigma$, a snapshot merge from agents $1, \dots, N$ is a diagram of local memory objects plus alignment maps induced by the place-identity relation. The consensus object $\tilde{\mathcal{K}}_t^i$ is the colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This reading gives three useful constraints.

1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an artefact of processing agent snapshots in a particular order.
2. **Chaining is expected.** If concept A aligns with B and B aligns with C , union-find performs the transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the chosen equivalence relation.
3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment morphisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean interpretation of broadcast snapshots as one diagram.

The model does *not* answer which perceptual features should identify two places. In the current grid substrate, identity still depends partly on coordinate keys and semantic tag similarity. Removing that coordinate anchor requires a separate place-identity experiment: agents must infer shared objects from local signatures rather than from global (x, y) keys.

7 Limitations and next experiments

Grid substrate. The current result is intentionally controlled: all experiments are in grid worlds. A continuous-state substrate such as VMAS, Habitat, ProcTHOR, or MiniWorld is the next portability step.

Coordinate anchor. The present implementation can use shared coordinates for atomic place keys. A stronger version of the paper should remove that anchor and add local signatures for place identity. The categorical merge story remains valid only after the alignment relation has been defined; it does not define the relation for us.

Minimal CSM. The CSM rules are deliberately simple. Adaptive cadence K_t , intent-conditioned trust, and selective consolidation of under-represented tags are the next architecture variants.

Learned baselines. CommNet-style learned communication is a useful comparison, but the more relevant future baseline is an explicit memory LLM or vector-memory multi-agent system measured on the same task.

8 Conclusion

Collective memory is not monotone. In scarce multi-agent navigation, instant information sharing can reduce completion by causing agents to converge on the same targets. Peer cadence exposes the trade-off, but cadence alone is a one-dimensional control surface. A collective semantic memory should make merge, trust, and staleness explicit. The minimal CSM does exactly that and moves the system to a new operating point: neither fast-broadcast M-saturation nor slow-broadcast C-saturation, but a more balanced $M \times C$ profile with higher success than every fixed peer cadence in the scarcity benchmark.

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341 **A Reproducibility commands**

- 342 • Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- 343
- 344 • Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- 345
- 346 • Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- 347
- 348
- 349 • Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- 350
- 351 • MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`
- 352

353 **B Asset and license note**

354 The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented
355 benchmark and memory code will be released under the MIT License as part of the artifact package.

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